

PROFESSIONAL EDUCATION

Digital Plus+ Programs

Smart Manufacturing:

Moving From Static to Dynamic Manufacturing Operations



Today's manufacturers have to approach a factory not just as a physical place, but also as a data center. Seeing physical operation and data as interconnected can improve quality, drive down costs, and increase the rate of production.

Dr. Brian W. Anthony

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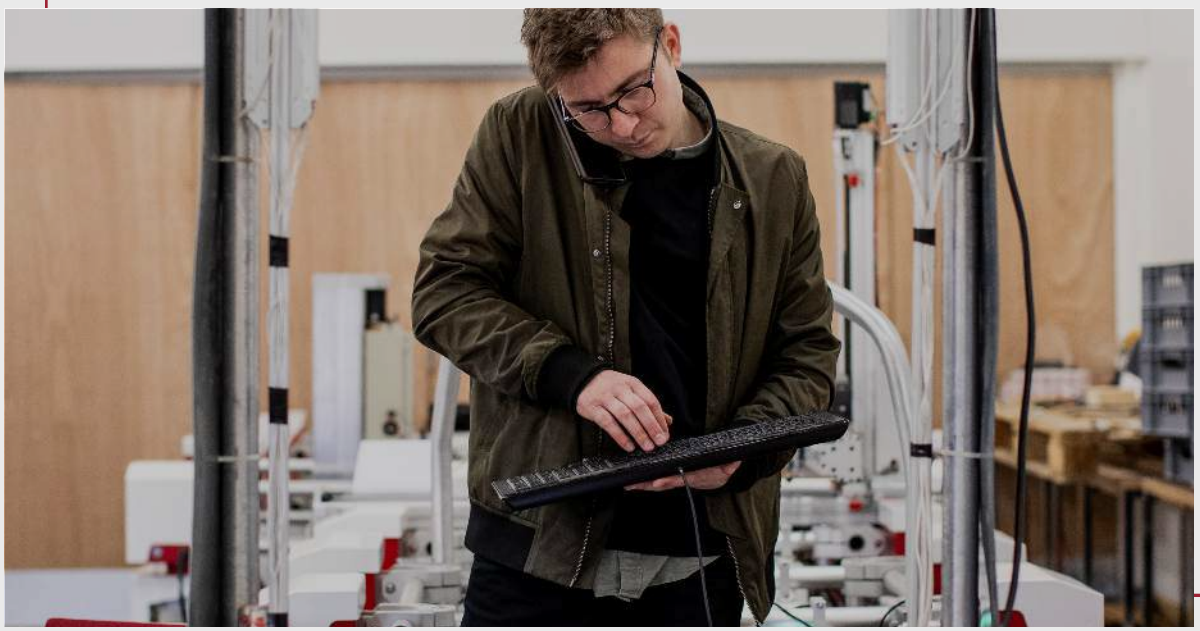
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The Role of Smart Manufacturing in Today's Landscape



Welcome to Industry 4.0, where new technologies and their rapid evolution are leading us to a **new industrial revolution based on cyberphysical production systems** that challenge traditional methods of carrying out operations in the manufacturing industry and make it more dynamic.

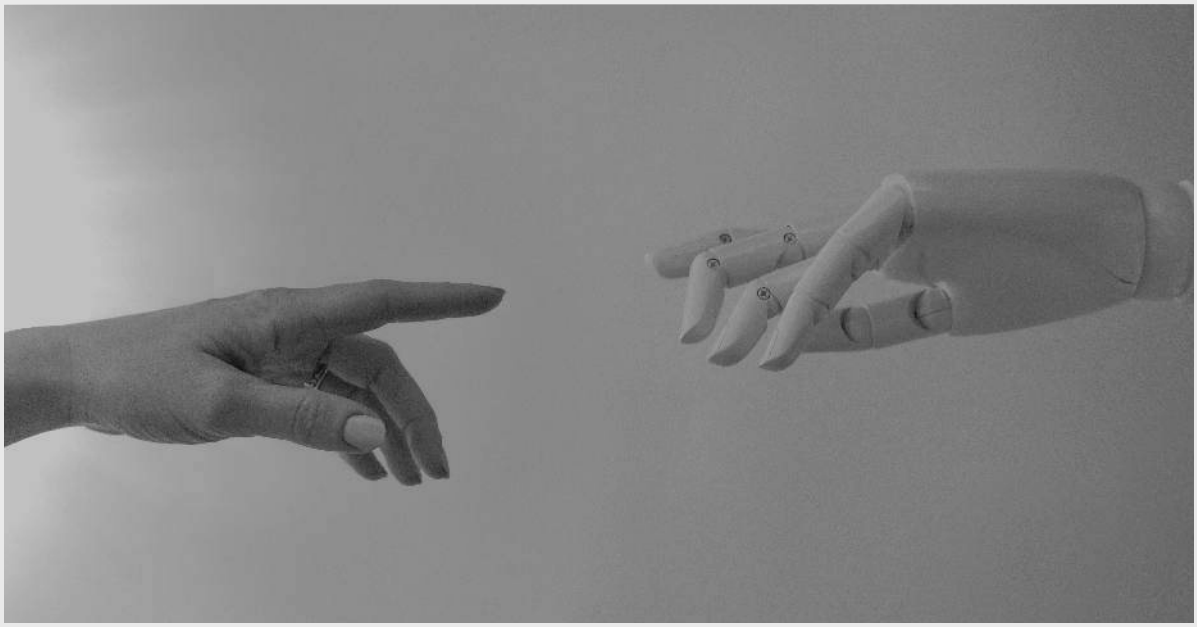
Smart manufacturing is a convergence of modern **data science** techniques and **artificial intelligence** to form the **factory of the future.**

The factory of the future is here

History is being made before our eyes as the future has already arrived. Get and stay ahead of this transformation with integral knowledge of **data analytics, sensors, software, and systems strategies**. Become a leader of the 4th Industrial Revolution. Start applying sweeping changes to improve your approach to the new manufacturing industry with Smart Manufacturing and **turn your organization into a Smart Enterprise** and propel the future forwards.

Optimize technology and take advantage of MIT Professional Education's **knowledge, expertise, and hands-on approach**. Drive your professional trajectory at full speed into the future of industry.

Smart manufacturing is about **increasing efficiency and eliminating pain points** in the system. It's characterized by a **highly connected,knowledge-enabled industrial enterprise** where all organizations and operating systems are linked, leading to enhanced productivity, sustainability, and economic performance.



About this Course

Optimize technology and take advantage of MIT Professional Education's knowledge, expertise, and hands-on approach. Drive your professional trajectory at full speed into the future of industry. **Smart manufacturing** is about increasing efficiency and eliminating pain points in the system. It's characterized by a highly connected, knowledge-enabled industrial enterprise where all organizations an operating systems are linked, leading to enhanced productivity, sustainability, and economic performance.



In keeping with MIT's founding principle Mens et Manus (Mind and Hand)—the synergy of theory and practice being at the heart of the learning experience— participants will learn by doing.

The course **centers around a smart machine**—a fiber extrusion device, fondly referred to as **FrED**—to demonstrate concepts as well as incorporate problemsolving skills into the curriculum. Furthermore, participants will have a chance to engage with their peers and expert learning facilitators to explore how these concepts can be leveraged in their organization.

10 weeks | 8-10 hours a week | Online

Key Takeaways

01 **Visualize the production of data** to better understand the key variables and phenomena underlying manufacturing.

02 **Control of the manufacturing processes,** feedback, process modeling, and monitoring.

03 **Application of advanced data analysis** to prepare statistic models and test manufacturing processes.

04 **Development of models** to make sense of data, and to examine and improve FrED.

05 **Discovery of the role played by sensors** in data production and smart manufacturing, and evaluation of the types of data that can be produced.

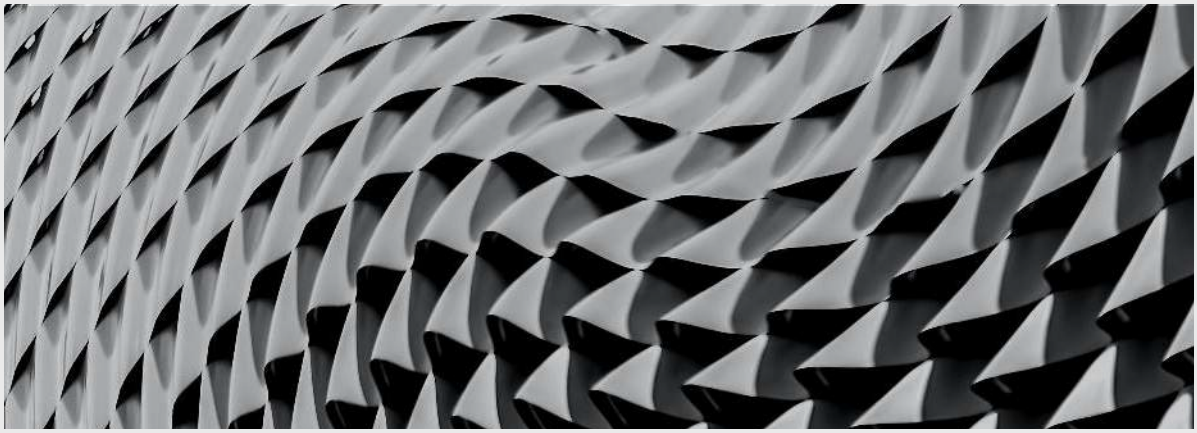
06 **Evaluation of sensors** and their respective data.

07 **Analysis of the differences** between the actual and predicted dynamics.

08 **Deepen understanding in innovation** of machines and artificial vision. Learn more about the different iterations of FrED, whose objective is to provide examples of optimizing machines.

09 **Examination of the discovery process** of data that best adjust to a model. The adaptation of models and sensitivity analysis establishes the connection between artificial vision and the control of statistical processes.

10 **Completion of an advanced data analysis** using datasets produced by FrED during the manufacturing process.



Participant Profile

This course is somewhat technical in nature, however this course material is **highly accessible for those new to smart manufacturing concepts**, while also being valuable for those who already have some experience. There are no prerequisites for this course. It is designed for:

Plants, production and operations managers who work in the manufacturing sector.

Data scientists who want to put their capabilities into practice in the field of smart manufacturing.

Design and manufacturing engineers who are looking to learn about data and development models in the manufacturing sector.

Consultants whose objective is to put additional value on the latest innovative technology in the manufacturing sector.

It is highly recommended that functional and cross-functional teams carry out the course together in order to accelerate the process of adopting smart manufacturing practices.

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Instructors



Dr. Brian W. Anthony

Director of the MIT Master of Engineering in Advanced Manufacturing and Design. Associate Director, MIT.nano

Dr. Anthony is the director of the MIT Master of Engineering in Advanced Manufacturing & Design, and codirector of the Medical Electronic Device Realization Center. He has more than 20 years of experience in product realization and is an **Emmy award winner** in the Academy of Television Arts & Sciences for his innovation in broadcast technology.

Dr. Anthony designs instruments and techniques to monitor and control physical systems. He also uses mechanical, electrical, and optical engineering as well as computer science and optimization to develop solutions.
Dr. Anthony's research revolves around the design of instruments and techniques for measuring and controlling complex physical systems. It includes the development of instruments and measurement solutions for manufacturing, medical diagnostics, and imaging diagnostics. In addition to his academic work, he has **extensive experience in market-driven technological innovation, product development and entrepreneurship, and marketing** at the point where information technology and advanced manufacturing come into contact. As an educator, his main interests are the modeling of large-scale systems dealing with decision-making and the development of optimization algorithms and software to analyze and design these systems. He is an expert in market-driven technological innovation as well as in entrepreneurship.

Dr. Anthony holds the following degrees:
• PhD in Engineering, Massachusetts Institute of Technology
• MS in Engineering, Massachusetts Institute of Technology
• BS in Engineering, Carnegie Mellon University broadcast technical innovation.

"Using technology to refine the manufacturing process has been a steady interest throughout my career and allows better products to be offered on the market, faster and more economically."

Course Outline

Module 1
Introduction to Smart Manufacturing and FrED Objectives

1. Global Trends
2. Data and its Importance
3. Smart Manufacturing
4. Fiber Extrusion Device (FrED)

Module 2
Analyzing Data: A Visualization Approach

1. Time-series analysis
2. Relationships in Time-Series plots
3. Moving Average
4. Investigating Periodicity
5. Tools of visualization

Module 3
Analyzing Data: A Visualization Approach

1. Extracting Periodicities Using Sine and Cosine Waves
2. Matching Signals and Frequency Representations
3. Data Visualization Tools
4. Exploring the Blocks on FrED
5. Speed of Response between Inputs and Outputs of FrED
6. Process Model of FrED
7. Connecting the Blocks
8. Making Sense of FrED's Data Using a Physical Model
9. Missing Elements in FrED's Model
10. Improving FrED
11. Summarizing FrED's Mode

Module 4
Sensors

1. Introduction to FrED's Sensors
2. Selecting a Sensor for FrED
3. Characteristics of Sensors
4. Time and Amplitude Discretization

Module 5
Control of Manufacturing Processes

1. Flow and Variation
2. Manufacturing Goals for Process Control
3. Simplifying the Block Diagrams
4. Models and Feedback Control
5. Control System Design
6. Impulse Response and Step Response
7. Modeling FrED I
8. Dynamic Response
9. Comparing Predicted and Actual Dynamics of FrED

Module 6
Machine Vision

1. Machine Vision and Motion Analysis
2. Images and Videos as Matrices
3. Pre-processing Images Using Histogram
4. Edge Detection
5. Image Filtering
6. Image Smoothing and Sharpening
7. Image Simplification by Morphological Operations
8. Image Interpretation
9. Image as a Measurement Tool

Module 7
Applications of Machine Vision

1. Using Cameras in the Manufacturing Environment
2. Manufacturing Techniques Applied to Sports
3. Cameras in Industrial Application
4. Smart Manufacturing Applied to Medicine

Module 8
Model Fitting and Sensitivity Analysis

1. Fiber Deflection Measurement
2. Model Fitting: An Overview
3. Applying Model Fitting and Sensitivity Analysis to Production
4. Model Fitting: A Recap

Module 9
Statistical Process Control

1. Statistics and Data
2. Sampling Measurement Error
3. Histograms and Data Distributions
4. Statistical Quality Control
5. Process Capability of Fred
6. Control Chart
7. Patterns in Control Charts
8. Image Interpretation
9. Control Charts on FrED

Module 10
Advanced Data Analysis

1. Correlation Coefficient
2. Cross-correlation Function
3. Cross-correlation Analysis
4. Production Data: Visual Analysis
5. Machine Learning Data Methods
6. FrED's Digital Twin
7. Smart Manufacturing: Tying It All Together

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Certificate of Completion

All participants who successfully complete **Smart Manufacturing: Moving From Static to Dynamic Manufacturing Operations** will receive an MIT Professional Education Certificate of Completion. In addition, they will also earn **8 Continuing Education Units (MIT CEUs).**

In order to obtain CEUs, participants must complete a required CEU confirmation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

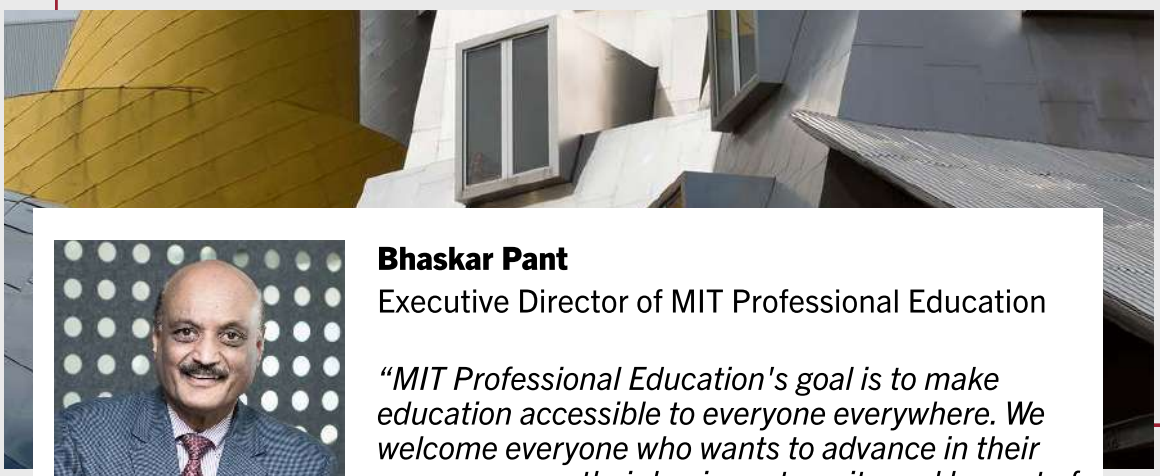
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

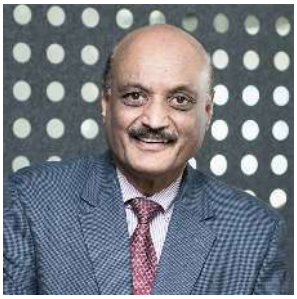
MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

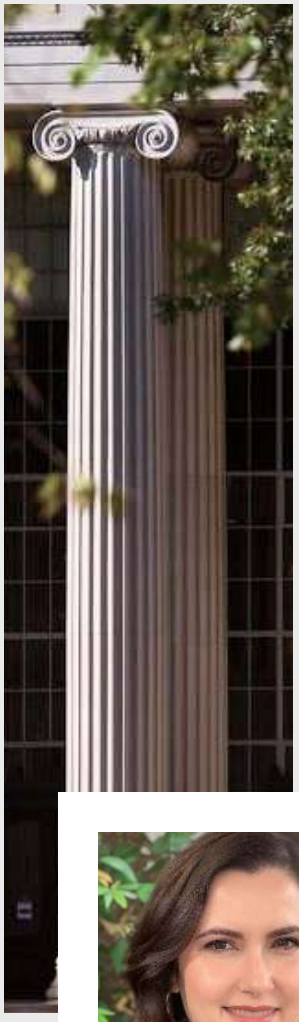


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

Industry 4.0 is transforming our world just as the Industrial Revolution transformed the world of yesterday. So much has evolved in recent years, but we are only just scratching the surface of what is possible and what is on the horizon with regards to technology, society, and intelligent manufacturing systems.

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Alumni.

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PROFESSIONAL EDUCATION
Digital Plus+ Programs

Are you ready to discover the factory of the future?

Frequently Asked Questions

global alumni
-excellence or nothing-

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